BIRDS CLASSIFICATION USING VOICE DATA

Abstract:

In this project, we developed an automated bird species classification system using voice data, with the aim of supporting ecological research and biodiversity monitoring. We collected a publicly available dataset of bird calls and songs, which included multiple species with varying audio patterns. The raw audio data was preprocessed by applying techniques such as noise reduction, segmentation, and normalization to enhance the quality of recordings.

Feature extraction was performed using Mel Frequency Cepstral Coefficients (MFCC), which effectively capture the timbral and tonal qualities of bird vocalizations. These features were then used to train a Convolutional Neural Network (CNN), selected for its strength in learning spatial hierarchies in spectrogram-like data.

The trained model demonstrated strong classification performance, achieving an accuracy of 97% on the test set. Our system can identify bird species based solely on their vocal patterns, making it a useful tool for non-invasive wildlife monitoring. The project highlights the potential of deep learning in ecological applications and sets the foundation for future real-time bird sound recognition systems.

Introduction

Birds play a vital role in maintaining ecological balance. Monitoring bird species is crucial for biodiversity conservation, and traditional methods involving manual identification are time-consuming and require expert knowledge. With the advancement of machine learning and signal processing, bird classification using voice data has emerged as a promising solution.

This project focuses on the development of a system that can automatically classify bird species based on their vocalizations. By analyzing audio recordings of bird calls and songs, the system extracts distinguishing acoustic features and applies classification algorithms to identify different species. The approach leverages techniques such as Mel Frequency Cepstral Coefficients (MFCC), spectrogram analysis, and machine learning models like Support Vector Machines (SVM), Convolutional Neural Networks (CNN), or Recurrent Neural Networks (RNN).

The goal of this project is to create an efficient, scalable, and accurate bird sound classification system that can aid ornithologists, researchers, and conservationists in real-time bird monitoring, especially in remote or densely vegetated. areas where visual identification is difficult.

Problem statement:

Identifying bird species manually is time-consuming and requires expert knowledge. An automated system that can classify birds based on their unique sounds using AI and ML.

Existing System:

Traditional bird species identification is manual, requiring expert knowledge and field observation, which is time-consuming, error-prone, and not scalable. Basic machine learning models used earlier had limited accuracy and robustness.

Proposed System:

The proposed system uses deep learning (CNN) and extracted features like MFCC from bird sounds to automate classification, achieving 97% accuracy, enabling fast, reliable, and scalable bird monitoring.

Objectives

- To develop a system that can identify bird species based on their sounds
- To collect and organize voice data of different bird species.
- To clean and process the audio data for better quality and analysis.
- To extract important features from the bird sounds using techniques like MFCC.
- To train machine learning or deep learning models for sound classification.
- To evaluate the performance of the model using accuracy and other metrics.
- To make bird identification faster and easier for researchers and nature lovers.

Methodology

1. Data Collection: Bird call datasets (Xeno-Canto)
2. Preprocessing: Noise reduction, spectrogram conversion
3. Feature Extraction: MFCC (Mel-Frequency Cepstral Coefficients), Chroma, Zero Crossing Rate (ZCR)
4. Modelling: CNN, Support vector machine (SVM), RF5.
5. Evaluation: Accuracy, Precision, Recall

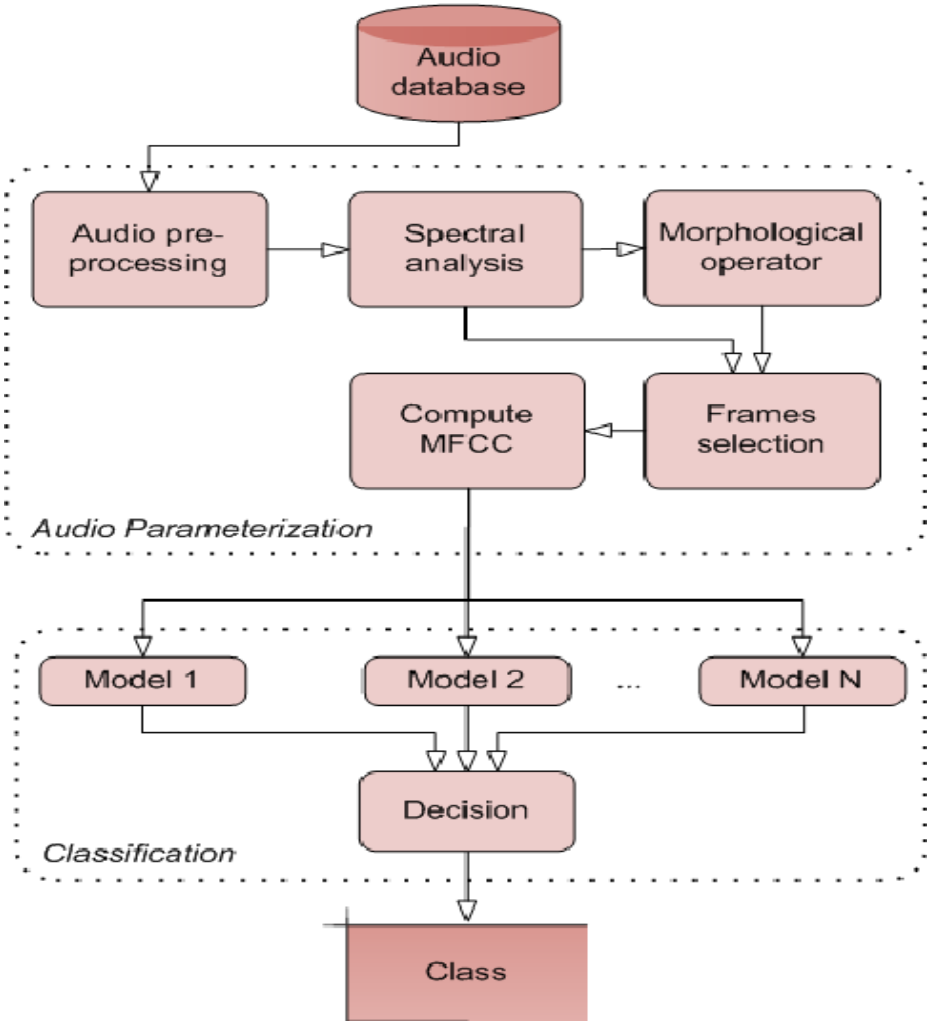

Fig.1 Methodology

Architecture

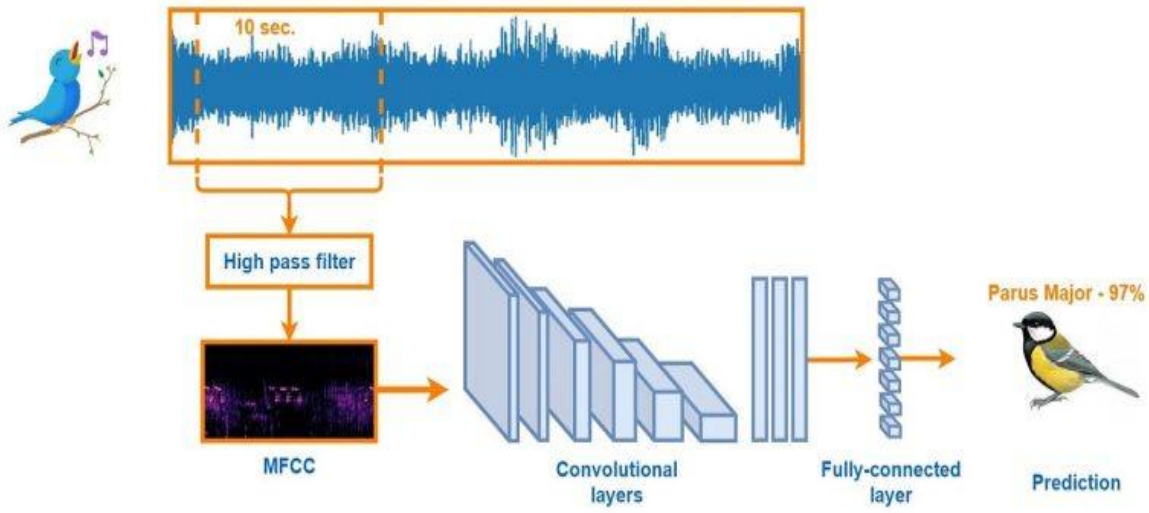

Fig .2 Architecture

Audio Input -> Preprocessing -> Feature Extraction -> Model Prediction -> Output Species Name

Software Requirements:

- Python (Programming language for model development)
- Jupyter Notebook / Google Colab (For coding and testing)
- Librosa (For audio processing and feature extraction)
- NumPy, Pandas (For data handling)
- Matplotlib / Seaborn (For visualizations)
- TensorFlow, Keras, PyTorch (For deep learning model development)
- Scikit-learn (For machine learning algorithms)

Hardware Requirements:

- Computer/Laptop with minimum 4 GB RAM (8 GB or more recommended)
- Processor: Intel i5 or equivalent (GPU support for faster training is optional)
- Microphone (if you plan to test real-time sound input)